

# 1 Continuous-Time ESKF Error-State Derivation

## 1.1 Nominal state and IMU motion model

Define the nominal state

$$x = [p \quad v \quad R \quad b_g \quad b_a \quad g]^T.$$

The continuous-time IMU motion equations are

$$\dot{p} = v, \tag{1.1}$$

$$\dot{v} = R(\tilde{a} - b_a - \eta_a) + g, \tag{1.2}$$

$$\dot{R} = R(\tilde{\omega} - b_g - \eta_g)^\wedge, \tag{1.3}$$

$$\dot{b}_g = \eta_{bg}, \tag{1.4}$$

$$\dot{b}_a = \eta_{ba}, \tag{1.5}$$

$$\dot{g} = 0. \tag{1.6}$$

The IMU measurement equations may be written as

$$\tilde{a} = R^T(a - g) + b_a + \eta_a, \tag{1.7}$$

$$\tilde{\omega} = \omega + b_g + \eta_g, \tag{1.8}$$

together with

$$\dot{R} = R\omega^\wedge.$$

For the nominal state, the kinematic equations have the same form but ignore the measurement-noise terms:

$$\dot{p} = v, \tag{1.9}$$

$$\dot{v} = R(\tilde{a} - b_a) + g, \tag{1.10}$$

$$\dot{R} = R(\tilde{\omega} - b_g)^\wedge, \tag{1.11}$$

$$\dot{b}_g = 0, \tag{1.12}$$

$$\dot{b}_a = 0, \tag{1.13}$$

$$\dot{g} = 0. \tag{1.14}$$

## 1.2 True state

Define the true state as

$$x_t = [p_t \quad v_t \quad R_t \quad b_{at} \quad b_{gt} \quad g_t]^T.$$

Its continuous-time dynamics are

$$\dot{p}_t = v_t, \tag{1.15}$$

$$\dot{v}_t = R_t(\tilde{a} - b_{at} - \eta_a) + g_t, \tag{1.16}$$

$$\dot{R}_t = R_t(\tilde{\omega} - b_{gt} - \eta_g)^\wedge, \tag{1.17}$$

$$\dot{b}_{gt} = \eta_{bg}, \tag{1.18}$$

$$\dot{b}_{at} = \eta_{ba}, \tag{1.19}$$

$$\dot{g}_t = 0. \tag{1.20}$$

### 1.3 Definition of the error state

Use a right-multiplicative rotation error:

$$p_t = p + \delta p, \quad (1.21)$$

$$v_t = v + \delta v, \quad (1.22)$$

$$R_t = R \delta R = R \text{Exp}(\delta \theta), \quad (1.23)$$

$$b_{gt} = b_g + \delta b_g, \quad (1.24)$$

$$b_{at} = b_a + \delta b_a, \quad (1.25)$$

$$g_t = g + \delta g. \quad (1.26)$$

The quantities without the subscript  $t$  are nominal-state variables. The goal is to derive the continuous-time dynamics of

$$\delta x = [\delta p \quad \delta v \quad \delta \theta \quad \delta b_g \quad \delta b_a \quad \delta g]^T.$$

### 1.4 Position error

Starting from

$$p_t = p + \delta p,$$

differentiate with respect to time:

$$\dot{p}_t = \dot{p} + \delta \dot{p}.$$

Since

$$\dot{p}_t = v_t = v + \delta v$$

and

$$\dot{p} = v,$$

we obtain

$$v + \delta v = v + \delta \dot{p}.$$

Therefore

$$\boxed{\delta \dot{p} = \delta v.} \quad (1.27)$$

### 1.5 Velocity error

Start from

$$v_t = v + \delta v.$$

Differentiating,

$$\dot{v}_t = \dot{v} + \delta \dot{v}.$$

Using the true-state equation,

$$\dot{v}_t = R_t(\tilde{a} - b_{at} - \eta_a) + g_t.$$

Substitute the error-state definitions:

$$\dot{v}_t = R \text{Exp}(\delta \theta)(\tilde{a} - b_a - \delta b_a - \eta_a) + g + \delta g.$$

For small  $\delta \theta$ ,

$$\text{Exp}(\delta \theta) = I + \delta \theta^\wedge + O(\|\delta \theta\|^2).$$

Keeping only first-order error terms,

$$\dot{v}_t \approx R(I + \delta\theta^\wedge)(\tilde{a} - b_a - \delta b_a - \eta_a) + g + \delta g \quad (1.28)$$

$$\approx R\tilde{a} - Rb_a - R\delta b_a - R\eta_a + R\delta\theta^\wedge\tilde{a} - R\delta\theta^\wedge b_a + g + \delta g. \quad (1.29)$$

Terms such as

$$\delta\theta^\wedge\delta b_a, \quad \delta\theta^\wedge\eta_a$$

are neglected as higher-order terms in this first-order error-state linearization.

Using

$$a^\wedge b = -b^\wedge a,$$

we have

$$\delta\theta^\wedge\tilde{a} = -\tilde{a}^\wedge\delta\theta, \quad -\delta\theta^\wedge b_a = b_a^\wedge\delta\theta.$$

Hence

$$\dot{v}_t \approx R(\tilde{a} - b_a) + g - R(\tilde{a} - b_a)^\wedge\delta\theta - R\delta b_a - R\eta_a + \delta g.$$

On the other hand, the nominal equation gives

$$\dot{v} + \delta\dot{v} = R(\tilde{a} - b_a) + g + \delta\dot{v}.$$

Comparing the two expressions,

$$\boxed{\delta\dot{v} = -R(\tilde{a} - b_a)^\wedge\delta\theta - R\delta b_a - \eta_a + \delta g.} \quad (1.30)$$

Here we use the standard ESKF noise assumption adopted in the text: multiplying the zero-mean white accelerometer noise by a rotation does not change its distribution. Thus, in distribution,

$$R\eta_a \sim \eta_a,$$

and the rotated noise term is written simply as

$$R\eta_a \Rightarrow \eta_a.$$

This is a distributional simplification rather than a pointwise identity between individual noise samples.

## 1.6 Rotation error

Start from

$$R_t = R \text{Exp}(\delta\theta).$$

Differentiating,

$$\dot{R}_t = \dot{R} \text{Exp}(\delta\theta) + R \frac{d}{dt} \text{Exp}(\delta\theta).$$

## Exact derivative of the exponential map

For a general time-varying

$$\delta\theta(t) \in \mathbb{R}^3,$$

the exact derivative is

$$\boxed{\frac{d}{dt} \text{Exp}(\delta\theta) = \text{Exp}(\delta\theta) \left( J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge}, \quad (1.31)$$

where  $J_r(\delta\theta)$  is the right Jacobian of  $\text{SO}(3)$ .

The formula above can be seen directly from the limit definition of the right Jacobian.

Define

$$Q(t) = \text{Exp}(\delta\theta(t)).$$

The right/local tangent velocity of  $Q(t)$  is

$$\omega_{\text{local}} = \lim_{h \rightarrow 0} \frac{\text{Log}(Q(t)^{-1} Q(t+h))}{h}.$$

Since

$$Q(t)^{-1} = \text{Exp}(-\delta\theta(t))$$

and

$$Q(t+h) = \text{Exp}(\delta\theta(t+h)),$$

we have

$$\omega_{\text{local}} = \lim_{h \rightarrow 0} \frac{\text{Log}[\text{Exp}(-\delta\theta(t)) \text{Exp}(\delta\theta(t+h))]}{h}.$$

Using the ordinary first-order expansion

$$\delta\theta(t+h) = \delta\theta(t) + h \delta\dot{\theta}(t) + O(h^2),$$

this becomes

$$\omega_{\text{local}} = \lim_{h \rightarrow 0} \frac{\text{Log}[\text{Exp}(-\delta\theta) \text{Exp}(\delta\theta + h \delta\dot{\theta})]}{h}.$$

Now compare this with the directional-limit definition of the right Jacobian:

$$\boxed{J_r(\phi)u = \lim_{h \rightarrow 0} \frac{\text{Log}[\text{Exp}(-\phi) \text{Exp}(\phi + hu)]}{h}}. \quad (1.32)$$

By identifying

$$\phi = \delta\theta, \quad u = \delta\dot{\theta},$$

we obtain

$$\boxed{\omega_{\text{local}} = J_r(\delta\theta) \delta\dot{\theta}}. \quad (1.33)$$

Since a right/local tangent velocity satisfies

$$\dot{Q} = Q \omega_{\text{local}}^\wedge,$$

we get

$$\dot{Q} = Q \left( J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge.$$

Substituting

$$Q = \text{Exp}(\delta\theta)$$

gives the exact relation

$$\boxed{\frac{d}{dt} \text{Exp}(\delta\theta) = \text{Exp}(\delta\theta) \left( J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge .} \quad (1.34)$$

Equivalently, this is the same relation as the compact right-Jacobian increment formula

$$\text{Exp}(\delta\theta + \delta\dot{\theta} dt) \approx \text{Exp}(\delta\theta) \text{Exp}\left( J_r(\delta\theta) \delta\dot{\theta} dt \right) .$$

To see why these two forms are equivalent, define

$$Q(t) = \text{Exp}(\delta\theta(t)).$$

Starting from the differential form,

$$\dot{Q} = Q \left( J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge ,$$

the ordinary first-order Taylor expansion in time gives

$$Q(t + dt) = Q(t) + \dot{Q}(t) dt + O(dt^2).$$

Substituting the differential equation,

$$Q(t + dt) = Q(t) + Q(t) \left( J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge dt + O(dt^2).$$

Factoring out  $Q(t)$ ,

$$Q(t + dt) = Q(t) \left[ I + \left( J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge dt \right] + O(dt^2).$$

For a small vector  $\epsilon$ ,

$$\text{Exp}(\epsilon) = I + \epsilon^\wedge + O(\|\epsilon\|^2).$$

Choose

$$\epsilon = J_r(\delta\theta) \delta\dot{\theta} dt.$$

Then

$$\text{Exp}\left( J_r(\delta\theta) \delta\dot{\theta} dt \right) = I + \left( J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge dt + O(dt^2).$$

Therefore

$$Q(t + dt) \approx Q(t) \text{Exp}\left( J_r(\delta\theta) \delta\dot{\theta} dt \right) .$$

Since

$$Q(t) = \text{Exp}(\delta\theta(t))$$

and

$$\delta\theta(t + dt) = \delta\theta(t) + \delta\dot{\theta}(t) dt + O(dt^2),$$

we obtain

$$\boxed{\text{Exp}(\delta\theta + \delta\dot{\theta} dt) \approx \text{Exp}(\delta\theta) \text{Exp}\left( J_r(\delta\theta) \delta\dot{\theta} dt \right) .} \quad (1.35)$$

Conversely, start from the increment form:

$$Q(t + dt) \approx Q(t) \text{Exp}\left(J_r(\delta\theta)\delta\dot{\theta} dt\right).$$

Using the small-angle expansion of the second exponential,

$$Q(t + dt) = Q(t) + Q(t) \left(J_r(\delta\theta)\delta\dot{\theta}\right)^\wedge dt + O(dt^2).$$

Hence

$$\frac{Q(t + dt) - Q(t)}{dt} = Q(t) \left(J_r(\delta\theta)\delta\dot{\theta}\right)^\wedge + O(dt).$$

Taking the limit  $dt \rightarrow 0$  gives

$$\boxed{\dot{Q} = Q \left(J_r(\delta\theta)\delta\dot{\theta}\right)^\wedge.} \quad (1.36)$$

Thus the increment form and the differential form describe the same local motion:

$$\boxed{\begin{aligned} \text{Exp}(\delta\theta + \delta\dot{\theta} dt) &\approx \text{Exp}(\delta\theta) \text{Exp}\left(J_r(\delta\theta)\delta\dot{\theta} dt\right) \\ &\Updownarrow dt \rightarrow 0 \\ \frac{d}{dt} \text{Exp}(\delta\theta) &= \text{Exp}(\delta\theta) \left(J_r(\delta\theta)\delta\dot{\theta}\right)^\wedge. \end{aligned}} \quad (1.37)$$

The upper expression is a first-order finite-increment relation, while the lower expression is the corresponding exact instantaneous differential relation.

Since

$$J_r(\delta\theta) = I - \frac{1}{2}\delta\theta^\wedge + O(\|\delta\theta\|^2),$$

the first-order ESKF approximation is

$$J_r(\delta\theta)\delta\dot{\theta} \approx \delta\dot{\theta}.$$

Therefore, within the first-order error-state derivation,

$$\boxed{\frac{d}{dt} \text{Exp}(\delta\theta) \approx \text{Exp}(\delta\theta)\delta\dot{\theta}^\wedge.} \quad (1.38)$$

Using this first-order approximation,

$$\dot{R}_t \approx \dot{R} \text{Exp}(\delta\theta) + R \text{Exp}(\delta\theta)\delta\dot{\theta}^\wedge.$$

The nominal rotation dynamics are

$$\dot{R} = R(\tilde{\omega} - b_g)^\wedge.$$

Hence

$$\dot{R}_t \approx R(\tilde{\omega} - b_g)^\wedge \text{Exp}(\delta\theta) + R \text{Exp}(\delta\theta)\delta\dot{\theta}^\wedge.$$

The true rotation dynamics are

$$\dot{R}_t = R_t(\tilde{\omega} - b_{gt} - \eta_g)^\wedge = R \text{Exp}(\delta\theta)(\tilde{\omega} - b_{gt} - \eta_g)^\wedge.$$

Equating both expressions and canceling  $R$  gives

$$\text{Exp}(\delta\theta)\delta\dot{\theta}^\wedge = \text{Exp}(\delta\theta)(\tilde{\omega} - b_{gt} - \eta_g)^\wedge \quad (1.39)$$

$$- (\tilde{\omega} - b_g)^\wedge \text{Exp}(\delta\theta). \quad (1.40)$$

Use the  $\text{SO}(3)$  identity

$$\phi^\wedge R = R(R^T \phi)^\wedge.$$

With

$$R = \text{Exp}(\delta\theta),$$

this gives

$$(\tilde{\omega} - b_g)^\wedge \text{Exp}(\delta\theta) = \text{Exp}(\delta\theta) (\text{Exp}(-\delta\theta)(\tilde{\omega} - b_g))^\wedge.$$

Therefore

$$\text{Exp}(\delta\theta)\delta\dot{\theta}^\wedge = \text{Exp}(\delta\theta) \left[ (\tilde{\omega} - b_{gt} - \eta_g)^\wedge \right. \quad (1.41)$$

$$\left. - (\text{Exp}(-\delta\theta)(\tilde{\omega} - b_g))^\wedge \right]. \quad (1.42)$$

For small  $\delta\theta$ ,

$$\text{Exp}(-\delta\theta) \approx I - \delta\theta^\wedge.$$

Thus

$$(\tilde{\omega} - b_{gt} - \eta_g) - \text{Exp}(-\delta\theta)(\tilde{\omega} - b_g) \quad (1.43)$$

$$\approx (\tilde{\omega} - b_{gt} - \eta_g) - (I - \delta\theta^\wedge)(\tilde{\omega} - b_g) \quad (1.44)$$

$$= b_g - b_{gt} - \eta_g + \delta\theta^\wedge \tilde{\omega} - \delta\theta^\wedge b_g. \quad (1.45)$$

Since

$$b_{gt} = b_g + \delta b_g,$$

we have

$$b_g - b_{gt} = -\delta b_g.$$

Also,

$$\delta\theta^\wedge(\tilde{\omega} - b_g) = -(\tilde{\omega} - b_g)^\wedge \delta\theta.$$

Therefore

$$\delta\dot{\theta} = -(\tilde{\omega} - b_g)^\wedge \delta\theta - \delta b_g - \eta_g.$$

Hence

$$\boxed{\delta\dot{\theta} = -(\tilde{\omega} - b_g)^\wedge \delta\theta - \delta b_g - \eta_g.} \quad (1.46)$$

### Identity used above

For  $R \in \text{SO}(3)$ ,

$$\text{Ad}_R \phi = (R\phi^\wedge R^T)^\vee = R\phi.$$

Thus

$$R\phi^\wedge R^T = (R\phi)^\wedge.$$

From this,

$$\phi^\wedge R = R(R^T \phi)^\wedge.$$

## 1.7 Supplement: From Finite Infinitesimal Increments to Instantaneous Velocity

This appendix clarifies how the right-Jacobian finite-increment relation is used to obtain the instantaneous right/local tangent velocity.

Let

$$Q(t) = \text{Exp}(\delta\theta(t)).$$

For a small time increment  $dt$ ,

$$\delta\theta(t + dt) = \delta\theta(t) + \delta\dot{\theta}(t) dt + O(dt^2),$$

and therefore

$$Q(t + dt) = \text{Exp}(\delta\theta(t + dt)) \approx \text{Exp}(\delta\theta + \delta\dot{\theta} dt).$$

The right-Jacobian increment relation is

$$\boxed{\text{Exp}(\phi + \Delta\phi) \approx \text{Exp}(\phi) \text{Exp}(J_r(\phi)\Delta\phi).} \quad (1.47)$$

Set

$$\phi = \delta\theta, \quad \Delta\phi = \delta\dot{\theta} dt.$$

Then

$$\boxed{Q(t + dt) \approx Q(t) \text{Exp}\left(J_r(\delta\theta)\delta\dot{\theta} dt\right).} \quad (1.48)$$

Now define the instantaneous right/local tangent velocity  $\omega_R$  by the infinitesimal motion

$$\boxed{Q(t + dt) \approx Q(t) \text{Exp}(\omega_R dt).} \quad (1.49)$$

Comparing the two infinitesimal expressions gives

$$\boxed{\omega_R = J_r(\delta\theta)\delta\dot{\theta}.} \quad (1.50)$$

Thus  $\delta\dot{\theta}$  is the time derivative of the exponential coordinates, whereas  $J_r(\delta\theta)\delta\dot{\theta}$  is the corresponding right/local tangent velocity of the rotation matrix.

To recover the ordinary matrix derivative, use

$$\text{Exp}(\omega_R dt) = I + \omega_R^\wedge dt + O(dt^2).$$

Hence

$$Q(t + dt) = Q(t) + Q(t)\omega_R^\wedge dt + O(dt^2),$$

so

$$\frac{Q(t + dt) - Q(t)}{dt} = Q(t)\omega_R^\wedge + O(dt).$$

Taking  $dt \rightarrow 0$ ,

$$\boxed{\dot{Q} = Q\omega_R^\wedge.} \quad (1.51)$$

Substituting

$$\omega_R = J_r(\delta\theta)\delta\dot{\theta}$$

and

$$Q = \text{Exp}(\delta\theta)$$

gives

$$\boxed{\frac{d}{dt} \text{Exp}(\delta\theta) = \text{Exp}(\delta\theta) \left( J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge.} \quad (1.52)$$

Therefore the finite infinitesimal increment and the instantaneous differential form are two descriptions of the same local motion:

$$\boxed{\begin{aligned} Q(t+dt) &\approx Q(t) \text{Exp}\left(J_r(\delta\theta) \delta\dot{\theta} dt\right) \\ &\Downarrow \quad dt \rightarrow 0 \\ \dot{Q} &= Q \left( J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge. \end{aligned}} \quad (1.53)$$

### Exact and small-angle forms of the right Jacobian

For  $\phi \in \mathbb{R}^3$ , let

$$\vartheta = \|\phi\|.$$

The exact right Jacobian of  $\text{SO}(3)$  is

$$\boxed{J_r(\phi) = I - \frac{1 - \cos \vartheta}{\vartheta^2} \phi^\wedge + \frac{\vartheta - \sin \vartheta}{\vartheta^3} (\phi^\wedge)^2.} \quad (1.54)$$

The coefficient of the first-order term follows from

$$\cos \vartheta = 1 - \frac{\vartheta^2}{2!} + \frac{\vartheta^4}{4!} - \cdots,$$

so

$$\frac{1 - \cos \vartheta}{\vartheta^2} = \frac{1}{2} - \frac{\vartheta^2}{24} + \cdots.$$

Similarly,

$$\sin \vartheta = \vartheta - \frac{\vartheta^3}{3!} + \frac{\vartheta^5}{5!} - \cdots,$$

which gives

$$\frac{\vartheta - \sin \vartheta}{\vartheta^3} = \frac{1}{6} - \frac{\vartheta^2}{120} + \cdots.$$

Therefore

$$J_r(\phi) = I - \frac{1}{2} \phi^\wedge + \frac{1}{6} (\phi^\wedge)^2 + O(\|\phi\|^3),$$

and, retaining only first-order terms,

$$\boxed{J_r(\phi) = I - \frac{1}{2} \phi^\wedge + O(\|\phi\|^2).} \quad (1.55)$$

For the ESKF rotation error,  $\phi = \delta\theta$  is small. Hence

$$J_r(\delta\theta) \delta\dot{\theta} = \delta\dot{\theta} - \frac{1}{2} \delta\theta^\wedge \delta\dot{\theta} + \cdots.$$

Under the first-order error-state approximation, the product involving  $\delta\theta$  and  $\delta\dot{\theta}$  is neglected, yielding

$$\boxed{J_r(\delta\theta) \delta\dot{\theta} \approx \delta\dot{\theta}.} \quad (1.56)$$

This is why the ESKF derivation commonly uses

$$\frac{d}{dt} \text{Exp}(\delta\theta) \approx \text{Exp}(\delta\theta) \delta\dot{\theta}^\wedge,$$

even though the exact differential relation contains the right Jacobian.

## 1.8 Gyroscope-bias error

Starting from

$$b_{gt} = b_g + \delta b_g,$$

differentiate:

$$\dot{b}_{gt} = \dot{b}_g + \delta \dot{b}_g.$$

The true bias dynamics are

$$\dot{b}_{gt} = \eta_{bg},$$

while the nominal model ignores noise:

$$\dot{b}_g = 0.$$

Therefore

$$\boxed{\delta \dot{b}_g = \eta_{bg}.} \tag{1.57}$$

## 1.9 Accelerometer-bias error

Similarly,

$$b_{at} = b_a + \delta b_a.$$

Differentiating,

$$\dot{b}_{at} = \dot{b}_a + \delta \dot{b}_a.$$

Using

$$\dot{b}_{at} = \eta_{ba}, \quad \dot{b}_a = 0,$$

we obtain

$$\boxed{\delta \dot{b}_a = \eta_{ba}.} \tag{1.58}$$

## 1.10 Gravity error

Starting from

$$g_t = g + \delta g,$$

differentiate:

$$\dot{g}_t = \dot{g} + \delta \dot{g}.$$

Since

$$\dot{g}_t = 0, \quad \dot{g} = 0,$$

we obtain

$$\boxed{\delta \dot{g} = 0.} \tag{1.59}$$

## 2 Final continuous-time error-state equations

Collecting the results,

$$\begin{aligned}
 \delta \dot{p} &= \delta v, \\
 \delta \dot{v} &= -R(\tilde{a} - b_a)^\wedge \delta \theta - R\delta b_a - \eta_a + \delta g, \\
 \delta \dot{\theta} &= -(\tilde{\omega} - b_g)^\wedge \delta \theta - \delta b_g - \eta_g, \\
 \delta \dot{b}_g &= \eta_{bg}, \\
 \delta \dot{b}_a &= \eta_{ba}, \\
 \delta \dot{g} &= 0.
 \end{aligned} \tag{2.1}$$

## 3 Discrete-Time ESKF Kinematic Equation

### 3.1 Nominal-state position equation

For the nominal state,

$$\dot{p} = v, \quad \dot{v} = R(\tilde{a} - b_a) + g.$$

Define the acceleration expressed in the world frame as

$$a_W(t) = R(t)(\tilde{a}(t) - b_a(t)) + g. \tag{3.1}$$

Since

$$\dot{p}(\tau) = v(\tau),$$

integrating from  $t$  to  $t + \Delta t$  gives

$$p(t + \Delta t) = p(t) + \int_t^{t+\Delta t} v(\tau) d\tau.$$

The velocity at an intermediate time  $\tau$  is

$$v(\tau) = v(t) + \int_t^\tau a_W(s) ds.$$

Substituting this into the position integral,

$$p(t + \Delta t) = p(t) + \int_t^{t+\Delta t} \left[ v(t) + \int_t^\tau a_W(s) ds \right] d\tau.$$

### Assumption over one IMU sampling interval

To obtain the discrete-time equation used here, assume that the world-frame acceleration is approximately constant over the short IMU sampling interval:

$$a_W(s) \approx a_W(t), \quad s \in [t, t + \Delta t]. \tag{3.2}$$

Equivalently, over this interval we use the values at the beginning of the interval,

$$R(s) \approx R(t), \quad \tilde{a}(s) \approx \tilde{a}(t), \quad b_a(s) \approx b_a(t),$$

with gravity  $g$  treated as constant. Thus

$$a_W(s) \approx R(t)(\tilde{a}(t) - b_a(t)) + g.$$

Under this approximation,

$$\int_t^\tau a_W(s) ds \approx a_W(t)(\tau - t).$$

Therefore

$$p(t + \Delta t) = p(t) + \int_t^{t+\Delta t} [v(t) + a_W(t)(\tau - t)] d\tau \quad (3.3)$$

$$= p(t) + v(t)\Delta t + a_W(t) \int_t^{t+\Delta t} (\tau - t) d\tau. \quad (3.4)$$

The remaining integral is

$$\int_t^{t+\Delta t} (\tau - t) d\tau = \frac{1}{2}\Delta t^2.$$

Hence

$$p(t + \Delta t) = p(t) + v(t)\Delta t + \frac{1}{2}a_W(t)\Delta t^2.$$

Substituting the expression for  $a_W(t)$ ,

$$p(t + \Delta t) = p(t) + v(t)\Delta t + \frac{1}{2}[R(t)(\tilde{a}(t) - b_a(t)) + g]\Delta t^2.$$

Finally,

$$\boxed{p(t + \Delta t) = p(t) + v(t)\Delta t + \frac{1}{2}R(t)(\tilde{a}(t) - b_a(t))\Delta t^2 + \frac{1}{2}g\Delta t^2.} \quad (3.5)$$

This is the discrete-time nominal position propagation equation. The appearance of  $R(t)$  rather than  $R(t + \Delta t)$  follows from the beginning-of-interval approximation above. In the exact continuous motion,  $R(s)$  and therefore  $a_W(s)$  generally vary throughout the sampling interval.

### 3.2 Nominal-state velocity equation

Starting from

$$\dot{v} = R(\tilde{a} - b_a) + g \equiv a_W,$$

integrate over the interval  $[t, t + \Delta t]$ :

$$v(t + \Delta t) = v(t) + \int_t^{t+\Delta t} a_W(\tau) d\tau.$$

Using the same assumption as in the position derivation,

$$a_W(\tau) \approx a_W(t), \quad \tau \in [t, t + \Delta t],$$

we obtain

$$v(t + \Delta t) \approx v(t) + a_W(t) \int_t^{t+\Delta t} d\tau.$$

Therefore

$$v(t + \Delta t) = v(t) + a_W(t)\Delta t.$$

Substituting

$$a_W(t) = R(t)(\tilde{a}(t) - b_a(t)) + g,$$

gives

$$\boxed{v(t + \Delta t) = v(t) + R(t)(\tilde{a}(t) - b_a(t))\Delta t + g\Delta t.} \quad (3.6)$$

This is the discrete-time nominal velocity propagation equation. As with the position equation, the approximation uses the beginning-of- interval values  $R(t)$ ,  $\tilde{a}(t)$ , and  $b_a(t)$  over one IMU sampling interval.

### 3.3 Nominal-state rotation equation

The continuous-time nominal rotation dynamics are

$$\boxed{\dot{R}(t) = R(t)(\tilde{\omega}(t) - b_g(t))^\wedge.} \quad (3.7)$$

Define the bias-corrected body-frame angular velocity

$$\omega_B(t) \equiv \tilde{\omega}(t) - b_g(t).$$

Over one short IMU sampling interval, assume

$$\boxed{\omega_B(\tau) \approx \omega_B(t), \quad \tau \in [t, t + \Delta t].} \quad (3.8)$$

Equivalently,

$$\tilde{\omega}(\tau) \approx \tilde{\omega}(t), \quad b_g(\tau) \approx b_g(t)$$

over this interval. The differential equation therefore becomes

$$\dot{R}(\tau) = R(\tau)\omega_B(t)^\wedge,$$

where  $\omega_B(t)^\wedge$  is constant with respect to  $\tau$ .

The solution of this matrix differential equation with initial condition  $R(t)$  is

$$R(\tau) = R(t) \exp(\omega_B(t)^\wedge(\tau - t)),$$

where  $\exp(\cdot)$  denotes the matrix exponential.

This can be verified directly:

$$\frac{d}{d\tau} R(\tau) = R(t) \frac{d}{d\tau} \exp(\omega_B(t)^\wedge(\tau - t)) \quad (3.9)$$

$$= R(t) \exp(\omega_B(t)^\wedge(\tau - t)) \omega_B(t)^\wedge \quad (3.10)$$

$$= R(\tau) \omega_B(t)^\wedge. \quad (3.11)$$

At  $\tau = t + \Delta t$ ,

$$R(t + \Delta t) = R(t) \exp(\omega_B(t)^\wedge \Delta t).$$

Since

$$\text{Exp}(\phi) \equiv \exp(\phi^\wedge)$$

and

$$(\omega_B \Delta t)^\wedge = \omega_B^\wedge \Delta t,$$

we obtain

$$R(t + \Delta t) = R(t) \text{Exp}(\omega_B(t)\Delta t).$$

Substituting

$$\omega_B(t) = \tilde{\omega}(t) - b_g(t),$$

gives the discrete-time nominal rotation propagation equation

$$\boxed{R(t + \Delta t) = R(t) \text{Exp}((\tilde{\omega}(t) - b_g(t))\Delta t).} \quad (3.12)$$

The exponential increment appears on the right because

$$\dot{R} = R\hat{\omega}_B$$

uses the body-frame (right-trivialized) angular velocity. In contrast, if the angular velocity were expressed in the world frame so that

$$\dot{R} = \omega_W^\wedge R,$$

the corresponding constant-angular-velocity propagation would be

$$R(t + \Delta t) = \text{Exp}(\omega_W\Delta t)R(t).$$

### 3.4 Nominal-state bias and gravity equations

For the nominal state, the gyroscope bias, accelerometer bias, and gravity are modeled as constant during propagation:

$$\dot{b}_g = 0, \quad \dot{b}_a = 0, \quad \dot{g} = 0.$$

Integrating the gyroscope-bias equation over one sampling interval,

$$b_g(t + \Delta t) = b_g(t) + \int_t^{t+\Delta t} \dot{b}_g(\tau) d\tau,$$

gives

$$\boxed{b_g(t + \Delta t) = b_g(t).} \quad (3.13)$$

Similarly,

$$b_a(t + \Delta t) = b_a(t) + \int_t^{t+\Delta t} \dot{b}_a(\tau) d\tau,$$

so

$$\boxed{b_a(t + \Delta t) = b_a(t).} \quad (3.14)$$

For gravity,

$$g(t + \Delta t) = g(t) + \int_t^{t+\Delta t} \dot{g}(\tau) d\tau,$$

and therefore

$$\boxed{g(t + \Delta t) = g(t).} \quad (3.15)$$

The constant nominal-bias equations do not imply that the physical IMU biases are assumed to remain exactly constant. Their random-walk variations are represented in the error-state dynamics by

$$\delta \dot{b}_g = \eta_{bg}, \quad \delta \dot{b}_a = \eta_{ba},$$

and their uncertainty is propagated through the error-state covariance.

### 3.5 Summary of the discrete-time nominal-state equations

Collecting the nominal-state propagation equations,

$$\begin{aligned}
p(t + \Delta t) &= p(t) + v(t)\Delta t + \frac{1}{2} [R(t)(\tilde{a}(t) - b_a(t)) + g(t)] \Delta t^2, \\
v(t + \Delta t) &= v(t) + [R(t)(\tilde{a}(t) - b_a(t)) + g(t)] \Delta t, \\
R(t + \Delta t) &= R(t) \text{Exp}((\tilde{\omega}(t) - b_g(t))\Delta t), \\
b_g(t + \Delta t) &= b_g(t), \\
b_a(t + \Delta t) &= b_a(t), \\
g(t + \Delta t) &= g(t).
\end{aligned} \tag{3.16}$$

### 3.6 Discrete-time error-state equations

We now discretize the continuous-time error-state equations derived previously.

#### Position error

From the continuous-time position-error equation,

$$\delta \dot{p} = \delta v,$$

assume that  $\delta v$  is approximately constant over one short sampling interval  $[t, t + \Delta t]$ . Then

$$\delta p(t + \Delta t) - \delta p(t) = \int_t^{t+\Delta t} \delta \dot{p}(\tau) d\tau \tag{3.17}$$

$$= \int_t^{t+\Delta t} \delta v(\tau) d\tau \tag{3.18}$$

$$\approx \delta v(t) \Delta t. \tag{3.19}$$

Therefore,

$$\boxed{\delta p(t + \Delta t) = \delta p(t) + \delta v(t) \Delta t.} \tag{3.20}$$

#### Velocity error

From the continuous-time velocity-error equation,

$$\delta \dot{v} = -R(\tilde{a} - b_a)^\wedge \delta \theta - R\delta b_a - \eta_a + \delta g,$$

define the deterministic part

$$\boxed{f_v = -R(\tilde{a} - b_a)^\wedge \delta \theta - R\delta b_a + \delta g.} \tag{3.21}$$

Then

$$\delta \dot{v} = f_v - \eta_a.$$

Integrating over one sampling interval,

$$\delta v(t + \Delta t) - \delta v(t) = \int_t^{t+\Delta t} f_v(\tau) d\tau - \int_t^{t+\Delta t} \eta_a(\tau) d\tau. \tag{3.22}$$

Assume that the deterministic quantities in  $f_v$  are approximately constant over the short interval:

$$\boxed{f_v(\tau) \approx f_v(t), \quad \tau \in [t, t + \Delta t].} \quad (3.23)$$

Thus

$$\int_t^{t+\Delta t} f_v(\tau) d\tau \approx f_v(t) \Delta t.$$

Define the integrated accelerometer-noise increment

$$\boxed{\eta_v \equiv \int_t^{t+\Delta t} \eta_a(\tau) d\tau.} \quad (3.24)$$

Assume the continuous-time accelerometer noise is zero-mean white Gaussian noise with

$$\mathbb{E}[\eta_a(t)] = 0$$

and

$$\boxed{\mathbb{E}[\eta_a(t)\eta_a(s)^T] = \sigma_a^2 I \delta(t - s).} \quad (3.25)$$

Equivalently,  $\sigma_a^2$  is the continuous-time white-noise intensity (per axis). Then the integrated noise increment also has zero mean:

$$\mathbb{E}[\eta_v] = 0.$$

Its covariance is

$$\text{Cov}(\eta_v) = \mathbb{E}[\eta_v \eta_v^T] \quad (3.26)$$

$$= \int_t^{t+\Delta t} \int_t^{t+\Delta t} \mathbb{E}[\eta_a(\tau)\eta_a(s)^T] d\tau ds \quad (3.27)$$

$$= \sigma_a^2 I \int_t^{t+\Delta t} \int_t^{t+\Delta t} \delta(\tau - s) d\tau ds \quad (3.28)$$

$$= \boxed{\sigma_a^2 \Delta t I.} \quad (3.29)$$

Thus, per axis,

$$\boxed{\text{Var}(\eta_v) = \sigma_a^2 \Delta t.} \quad (3.30)$$

The book distinguishes the sampled accelerometer measurement noise from the continuous-time noise-density parameter through

$$\boxed{\sigma_a(k) = \frac{\sigma_a}{\sqrt{\Delta t}.} \quad (3.31)$$

Therefore the standard deviation of the integrated velocity-noise increment can be written in either equivalent form:

$$\boxed{\sigma(\eta_v) = \Delta t \sigma_a(k) = \sigma_a \sqrt{\Delta t}.} \quad (3.32)$$

The first expression uses the sampled-noise notation, while the second follows directly after substituting the sampled-noise relation.

Note that writing simply

$$\text{Var}(\eta_a) = \sigma_a^2$$

for continuous-time white noise is shorthand for its noise intensity. The Dirac-delta covariance above makes the continuous-to-discrete scaling explicit.

Hence

$$\delta v(t + \Delta t) \approx \delta v(t) + f_v(t)\Delta t - \eta_v.$$

Substituting the expression for  $f_v(t)$  gives

$$\boxed{\begin{aligned} \delta v(t + \Delta t) = & \delta v(t) - R(t)(\tilde{a}(t) - b_a(t))^\wedge \delta \theta(t) \Delta t \\ & - R(t) \delta b_a(t) \Delta t + \delta g(t) \Delta t - \eta_v. \end{aligned}} \quad (3.33)$$

The term  $\eta_v$  is the discrete-time noise increment obtained by integrating the continuous-time accelerometer noise over the sampling interval; unlike the deterministic term  $f_v(t)\Delta t$ , it should not be interpreted simply as a deterministic constant multiplied by  $\Delta t$ .

### Rotation error

From the continuous-time rotation-error equation,

$$\delta \dot{\theta} = -(\tilde{\omega} - b_g)^\wedge \delta \theta - \delta b_g - \eta_g,$$

define

$$A \equiv -(\tilde{\omega} - b_g)^\wedge.$$

Over one short IMU sampling interval, assume that

$$\tilde{\omega}(\tau) \approx \tilde{\omega}(t), \quad b_g(\tau) \approx b_g(t),$$

so that  $A$  is approximately constant. The homogeneous equation

$$\dot{x} = Ax$$

has the solution

$$x(t + \Delta t) = \exp(A\Delta t)x(t).$$

Therefore the homogeneous part of the rotation error propagates as

$$\delta \theta(t + \Delta t) = \exp(-(\tilde{\omega}(t) - b_g(t))^\wedge \Delta t) \delta \theta(t).$$

Using the SO(3) exponential-map notation,

$$\exp(-(\tilde{\omega} - b_g)^\wedge \Delta t) = \text{Exp}(-(\tilde{\omega} - b_g)\Delta t).$$

For the forcing terms  $-\delta b_g - \eta_g$ , the first-order discretization used here gives

$$\boxed{\begin{aligned} \delta \theta(t + \Delta t) \approx & \text{Exp}(-(\tilde{\omega}(t) - b_g(t))^\wedge \Delta t) \delta \theta(t) \\ & - \delta b_g(t) \Delta t - \eta_\theta, \end{aligned}} \quad (3.34)$$

where the integrated gyroscope-noise increment is

$$\boxed{\eta_\theta \equiv \int_t^{t+\Delta t} \eta_g(\tau) d\tau.} \quad (3.35)$$

More exactly, for constant

$$A = -(\tilde{\omega} - b_g)^\wedge,$$

the variation-of-constants formula is

$$\delta \theta(t + \Delta t) = e^{A\Delta t} \delta \theta(t) - \int_t^{t+\Delta t} e^{A(t+\Delta t-\tau)} [\delta b_g(\tau) + \eta_g(\tau)] d\tau.$$

The expression above with  $-\delta b_g \Delta t - \eta_\theta$  is the corresponding first-order forcing-term approximation.

**Variance of the discrete rotation-noise increment.** Assume the continuous-time gyroscope noise is zero-mean isotropic white Gaussian noise with

$$\mathbb{E}[\eta_g(t)] = 0$$

and

$$\boxed{\mathbb{E}[\eta_g(t)\eta_g(s)^T] = \sigma_g^2 I \delta(t-s)}. \quad (3.36)$$

Then

$$\mathbb{E}[\eta_\theta] = 0$$

and

$$\text{Cov}(\eta_\theta) = \mathbb{E}[\eta_\theta \eta_\theta^T] \quad (3.37)$$

$$= \int_t^{t+\Delta t} \int_t^{t+\Delta t} \mathbb{E}[\eta_g(\tau)\eta_g(s)^T] d\tau ds \quad (3.38)$$

$$= \boxed{\sigma_g^2 \Delta t I}. \quad (3.39)$$

Thus, per axis,

$$\boxed{\text{Var}(\eta_\theta) = \sigma_g^2 \Delta t}. \quad (3.40)$$

For the sampled gyroscope measurement noise, the book uses

$$\boxed{\sigma_g(k) = \frac{\sigma_g}{\sqrt{\Delta t}}}. \quad (3.41)$$

Hence the standard deviation of the integrated rotation-noise increment can be written as

$$\boxed{\sigma(\eta_\theta) = \Delta t \sigma_g(k) = \sigma_g \sqrt{\Delta t}}. \quad (3.42)$$

Again, the first form uses the sampled-noise notation, while the second is the equivalent expression in terms of the continuous-time noise-density parameter. The  $\sqrt{\Delta t}$  factor belongs to the standard deviation, not to the variance. The same covariance result also holds if the exact state-transition rotation inside the noise integral is retained: because  $A$  is skew-symmetric,  $e^{As}$  is a rotation matrix, and for isotropic noise

$$e^{As}(\sigma_g^2 I)e^{AsT} = \sigma_g^2 I.$$

### Gyroscope-bias error

Here we explicitly distinguish the continuous-time bias random-walk process from its integrated discrete-time increment. The continuous process is denoted by

$$\eta_{bg}(t),$$

while

$$\eta_{bg,d}$$

denotes the noise accumulated over one sampling interval. The book reuses the symbol  $\eta_{bg}$  for the discrete increment in Eqs. (3.42)–(3.43); the subscript  $d$  is retained here to avoid ambiguity.

From

$$\delta \dot{b}_g = \eta_{bg},$$

integrating over one sampling interval gives

$$\delta b_g(t + \Delta t) = \delta b_g(t) + \int_t^{t+\Delta t} \eta_{bg}(\tau) d\tau.$$

Define

$$\eta_{bg,d} \equiv \int_t^{t+\Delta t} \eta_{bg}(\tau) d\tau. \quad (3.43)$$

Then

$$\delta b_g(t + \Delta t) = \delta b_g(t) + \eta_{bg,d}. \quad (3.44)$$

If

$$\mathbb{E}[\eta_{bg}(t)\eta_{bg}(s)^T] = \sigma_{bg}^2 I \delta(t - s),$$

then

$$\text{Cov}(\eta_{bg,d}) = \sigma_{bg}^2 \Delta t I, \quad (3.45)$$

or, per axis,

$$\text{Var}(\eta_{bg,d}) = \sigma_{bg}^2 \Delta t, \quad \sigma_{\eta_{bg,d}} = \sigma_{bg} \sqrt{\Delta t}. \quad (3.46)$$

### Accelerometer-bias error

Likewise,  $\eta_{ba}(t)$  denotes the continuous-time accelerometer-bias random-walk process, while  $\eta_{ba,d}$  denotes its integrated discrete-time increment over one sampling interval.

Similarly, from

$$\delta \dot{b}_a = \eta_{ba},$$

we obtain

$$\delta b_a(t + \Delta t) = \delta b_a(t) + \eta_{ba,d}, \quad (3.47)$$

where

$$\eta_{ba,d} \equiv \int_t^{t+\Delta t} \eta_{ba}(\tau) d\tau. \quad (3.48)$$

For continuous-time isotropic white noise intensity  $\sigma_{ba}^2$ ,

$$\text{Cov}(\eta_{ba,d}) = \sigma_{ba}^2 \Delta t I, \quad (3.49)$$

and therefore

$$\text{Var}(\eta_{ba,d}) = \sigma_{ba}^2 \Delta t, \quad \sigma_{\eta_{ba,d}} = \sigma_{ba} \sqrt{\Delta t}. \quad (3.50)$$

### Gravity error

From

$$\delta \dot{g} = 0,$$

integration gives

$$\delta g(t + \Delta t) - \delta g(t) = \int_t^{t+\Delta t} \delta \dot{g}(\tau) d\tau = 0.$$

Therefore

$$\delta g(t + \Delta t) = \delta g(t). \quad (3.51)$$

### 3.7 Summary of the discrete-time error-state equations

Collecting the discrete-time error-state propagation equations,

$$\begin{aligned}
\delta p_{k+1} &= \delta p_k + \delta v_k \Delta t, \\
\delta v_{k+1} &= \delta v_k - R_k(\tilde{a}_k - b_{a,k})^\wedge \delta \theta_k \Delta t - R_k \delta b_{a,k} \Delta t + \delta g_k \Delta t - \eta_v, \\
\delta \theta_{k+1} &\approx \text{Exp}(-(\tilde{\omega}_k - b_{g,k}) \Delta t) \delta \theta_k - \delta b_{g,k} \Delta t - \eta_\theta, \\
\delta b_{g,k+1} &= \delta b_{g,k} + \eta_{bg,d}, \\
\delta b_{a,k+1} &= \delta b_{a,k} + \eta_{ba,d}, \\
\delta g_{k+1} &= \delta g_k.
\end{aligned} \tag{3.52}$$

It is useful to list both the standard deviations and variances. The sampled IMU measurement-noise standard deviations satisfy

$$\sigma_a(k) = \frac{\sigma_a}{\sqrt{\Delta t}}, \quad \sigma_g(k) = \frac{\sigma_g}{\sqrt{\Delta t}}.$$

Therefore,

$$\begin{aligned}
\sigma(\eta_v) &= \Delta t \sigma_a(k) = \sigma_a \sqrt{\Delta t}, & \text{Var}(\eta_v) &= \sigma_a^2 \Delta t, \\
\sigma(\eta_\theta) &= \Delta t \sigma_g(k) = \sigma_g \sqrt{\Delta t}, & \text{Var}(\eta_\theta) &= \sigma_g^2 \Delta t, \\
\sigma(\eta_{bg,d}) &= \sigma_{bg} \sqrt{\Delta t}, & \text{Var}(\eta_{bg,d}) &= \sigma_{bg}^2 \Delta t, \\
\sigma(\eta_{ba,d}) &= \sigma_{ba} \sqrt{\Delta t}, & \text{Var}(\eta_{ba,d}) &= \sigma_{ba}^2 \Delta t.
\end{aligned} \tag{3.53}$$

The first two standard-deviation expressions use the sampled IMU-noise notation. For the bias random walks, the book denotes the integrated discrete noises again by  $\eta_{bg}$  and  $\eta_{ba}$ ; this note uses  $\eta_{bg,d}$  and  $\eta_{ba,d}$  so that they remain distinct from the continuous-time processes  $\eta_{bg}(t)$  and  $\eta_{ba}(t)$ .

## 4 Motion Process of ESKF

Using the discrete-time error-state equations derived above, write the error-state motion model as

$$\delta x_{k+1} = f(\delta x_k) + w_k, \quad w_k \sim \mathcal{N}(0, Q_k). \tag{4.1}$$

The discrete process-noise covariance is

$$Q_k = \text{diag}(0_3, \text{Cov}(\eta_v), \text{Cov}(\eta_\theta), \text{Cov}(\eta_{bg,d}), \text{Cov}(\eta_{ba,d}), 0_3). \tag{4.2}$$

Here the subscript  $d$  is retained for the bias-noise increments in order to distinguish them from the corresponding continuous-time white-noise processes.

### 4.1 Linearized form

Linearize the discrete transition function  $f$  about the nominal error-state mean. For a general linearization point  $\bar{\delta x}_k$ ,

$$f(\delta x_k) \approx f(\bar{\delta x}_k) + \left. \frac{\partial f(\delta x)}{\partial \delta x} \right|_{\delta x = \bar{\delta x}_k} (\delta x_k - \bar{\delta x}_k).$$

In the ESKF, the error-state mean is reset to zero, so the linearization point is

$$\bar{\delta x}_k = 0.$$

Therefore,

$$f(\delta x_k) \approx f(0) + \left. \frac{\partial f(\delta x)}{\partial \delta x} \right|_{\delta x=0} \delta x_k.$$

For the error-state motion model,

$$f(0) = 0.$$

Define the discrete-time error-state transition Jacobian as

$$F_k \triangleq \left. \frac{\partial f(\delta x)}{\partial \delta x} \right|_{\delta x=0}. \quad (4.3)$$

The linearized error-state propagation is therefore

$$\delta x_{k+1} \approx F_k \delta x_k + w_k. \quad (4.4)$$

Thus,  $F_k$  is the Jacobian of the *discrete-time error-state transition function* with respect to the error state, evaluated at zero error. It should not be confused with the time derivative  $d\delta x/dt$ .

## 4.2 State transition Jacobian matrix

The error-state ordering is

$$\delta x = [\delta p \quad \delta v \quad \delta \theta \quad \delta b_g \quad \delta b_a \quad \delta g]^T. \quad (4.5)$$

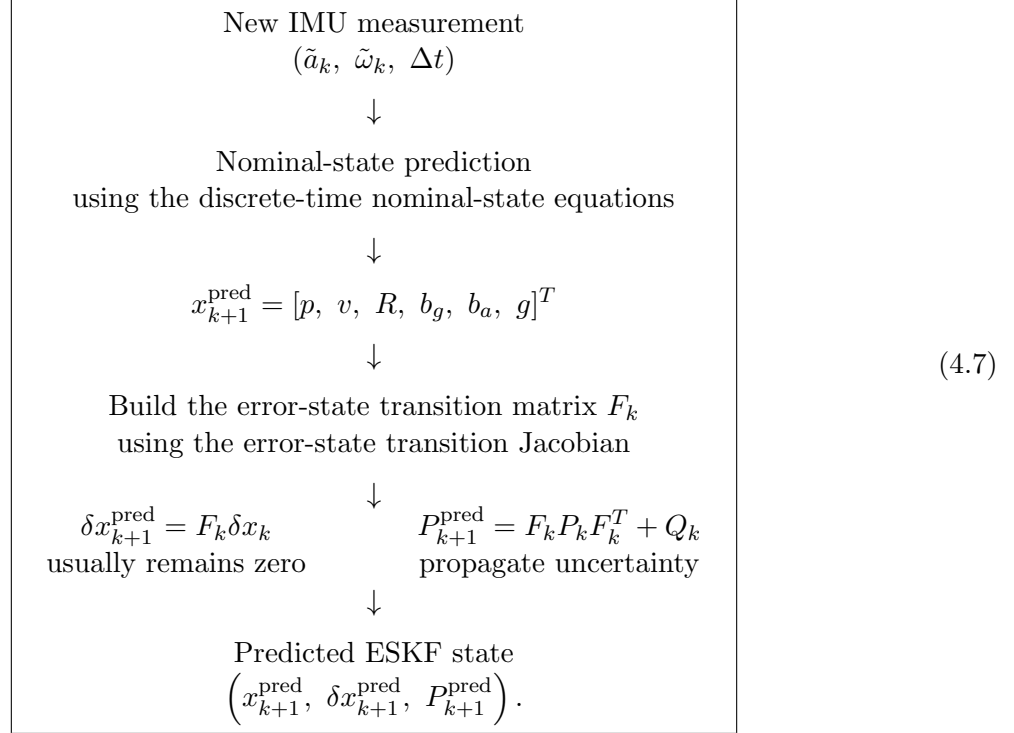
Since the discrete-time error-state equations have already been linearized explicitly, the transition Jacobian can be read directly from the coefficients multiplying each error-state component. With the ordering above,

$$F_k = \begin{bmatrix} I_3 & I_3 \Delta t & 0 & 0 & 0 & 0 \\ 0 & I_3 & -R(\tilde{a} - b_a)^\wedge \Delta t & 0 & -R \Delta t & I_3 \Delta t \\ 0 & 0 & \text{Exp}(-(\tilde{\omega} - b_g) \Delta t) & -I_3 \Delta t & 0 & 0 \\ 0 & 0 & 0 & I_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & I_3 \end{bmatrix}. \quad (4.6)$$

Each entry is a  $3 \times 3$  block, so  $F_k \in \mathbb{R}^{18 \times 18}$ .

### 4.3 ESKF IMU propagation flow

For each new IMU measurement, the ESKF prediction process can be summarized as follows:



Mathematically, the prediction contains three steps:

1. propagate the nominal state using the IMU measurements;
2. propagate the error-state mean,

$$\delta x_{k+1}^{\text{pred}} = F_k \delta x_k;$$

3. propagate the error-state covariance,

$$P_{k+1}^{\text{pred}} = F_k P_k F_k^T + Q_k.$$

In the ESKF, the error-state mean is reset to zero after the update step. Therefore, in the usual prediction cycle,

$$\delta x_k = 0 \implies \delta x_{k+1}^{\text{pred}} = F_k \delta x_k = 0.$$

Consequently, the explicit error-state mean propagation is often unnecessary in an implementation. The important quantities propagated during the IMU prediction are the nominal state and the error-state covariance.